



NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES
(KARACHI CAMPUS)
FAST School of Computing
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Project Proposal: [Color Hunt Game]

Abstract:

Color Hunt is a GUI-based interactive game developed in Java, demonstrating Object-Oriented Programming principles such as **Encapsulation**, **Inheritance**, **Polymorphism**, and **Abstraction**. The game challenges players to identify the **color in which a word is displayed** — not the word itself — enhancing reaction time, focus, and cognitive reflexes.

Key Features:

Real-time score tracking

Adaptive difficulty with increasing speed

Custom-designed GUI components for a polished experience

Sound effects to enhance interactivity

A built-in mini-game (Tic Tac Toe):

After 3 consecutive wrong answers, players enter a Tic Tac Toe match. Winning allows the main game to continue; losing ends the session — adding an innovative second-chance mechanism.

The game consists of **10+ classes** including

- **Game:** Controls the overall game flow and logic
- **UIManager:** Manages all GUI components and visual layout
- **ColorText:** Handles word-color mismatches
- **GameMode:** An abstract class defining the structure of game modes
- **ClassicMode & ChallengeMode:** Concrete classes representing gameplay variations
- **Scoreboard:** Dynamically tracks and updates the player's score
- **Player:** Stores high scores
- **SoundManager:** Enhances gameplay experience with audio feedback
- **TicTacToe:** Implements the mini-game for recovery after repeated mistakes
- **GameFlowManager:** Coordinates the main game loop and transitions between game states
- **GameException:** A custom exception class for game-specific error scenarios, improving robustness
- **Color Challenge:** An abstract class defining the structure of game modes
- **Match Color:** Implements the main gameplay logic where players match the displayed color of a word, not its text.

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